# RENNER ACADEMY 2025

## Voicing the Renner Hammerhead

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Creator: Jens- Sören Kruse | Klavierbaumeister | © RENNER ACADEMY

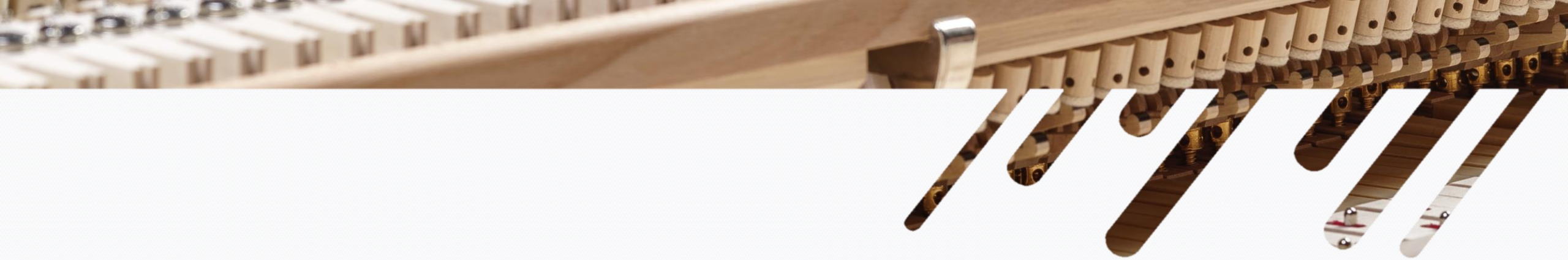


# Course topics

- Fundamentals of voicing
- Unchangeable challanges
- Challenges we can influence
- Regulation & fine tuning
- Working steps and working goals
- Principle of voicing areas | dynamic wedges
- Needle pricks grafik | stitch direction and result
- First step of pre-voicing | Pre-needleing
- Pre-voicing | clamping hammers | needleing
- Sanding | forming | ironing surface of hammerheads
- Setting strings on frame pin
- Setting strings on bridge pin
- String leveling
- Final travling | spuare | alignement of hammerheads
- Check strike point and straightness of hammerheads
- File hammers to strings
- Mid- voicing | find uneven sections and balance
- Final balance voicing
- Shift (left) padel voicing

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Quick Overview



# Fundamentals of voicing:

- Bringing the sound characteristics of an instrument to the level at which it sounds most beautiful
- All notes with as many dynamic levels as possible
- All tones with the widest possible partial tone spectrum

## Challange construction

- Different constructions
- Different materials
- Different sound philosophys
- Different manufacturing processes







**Challange location**

- Different rooms
- Different acoustics
- Different applications





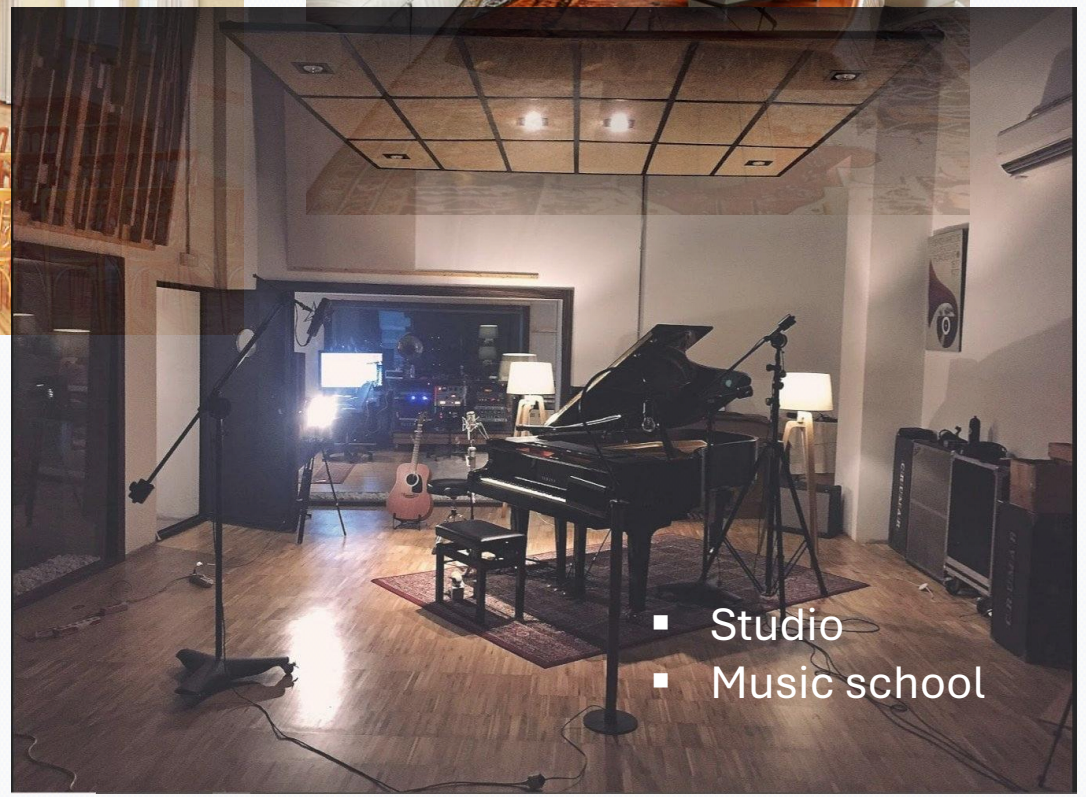
- Chamber music hall



- Living room



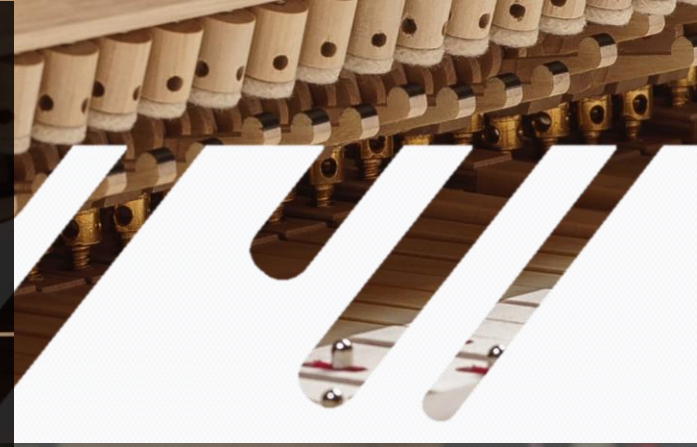
- Big concert hall



- Studio
- Music school







## Challange user

- Technical user for showroom
- Professional pianist
- Privat user
- Player (tuch & tone)
- Audience and concertgoers

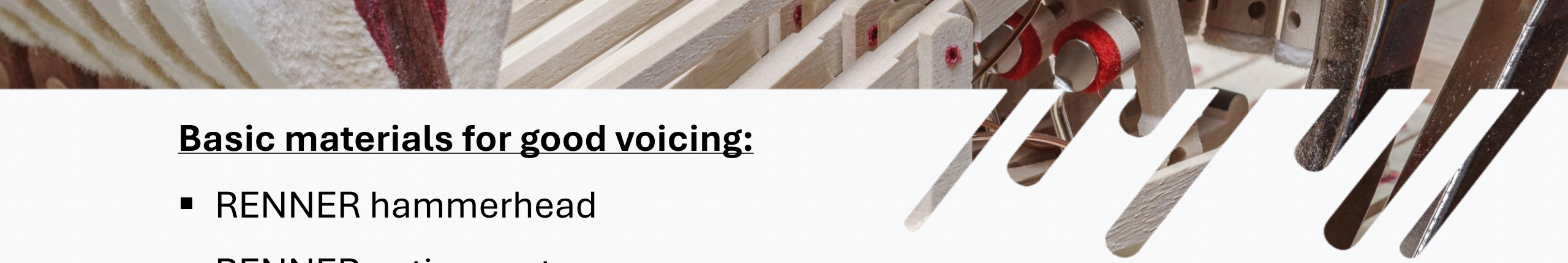






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Unchangeable challanges



**Basic materials for good voicing:**

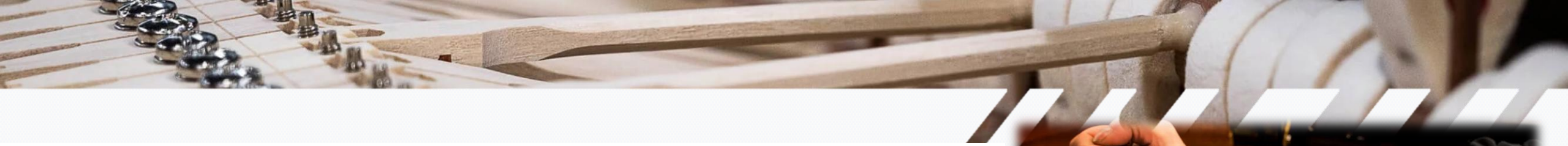
- RENNER hammerhead
- RENNER action parts

**Basic works of good voicing:**

- Perfect install of new hammershanks and gluing of the hammerheads
- Perfect cleaned instrument
- Perfect stringing
- Perfect regulating & tuning

**Training:**

- RENNER ACADEMY professional training



**Regulation of GP-Actions**

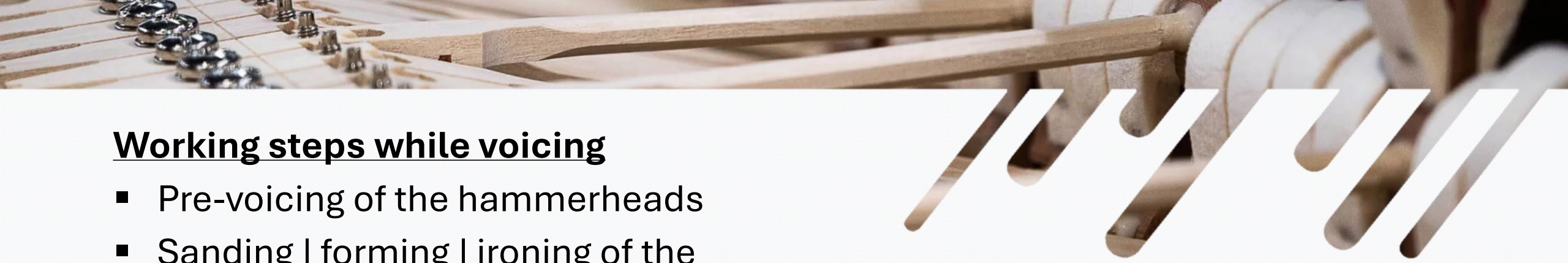
The following regulation sequence is just one of many possibilities.

- Cleaning, string spacing, rough tuning
- Sanding / levelling keybed
- Fitting of keyframe without shift spring
- Adjust key frame adjustment screws if available
- Key easing
- Check and correct jack position
- Setting action back on keyframe together with keys
- Adjustment of the optimal sound in treble section
- Alignment and travelling of hammers
- Alignment of whippens below hammer knuckles, capstans and let off screws
- Final check of alignment jack to knuckle
- Levelling the keys
- Set off
- Drop of hammers
- Trial keys for after touch and key depth and necessary hammer blow
- Hammer blow
- Key depth and after touch
- Re-check of hammer drop
- Bending back checks to hammer tails
- Back check adjustment
- Repetition spring adjustment
- Check shift pedal
- Check alignment damper timing
- Check damper half time (spoon bending)
- Damper stop rail



- No clean voicing without good regulation,
- and without good regulation, no clean tuning,
- and without clean tuning, no clean voicing.

- One leads to the other, and everything is connected with everything.

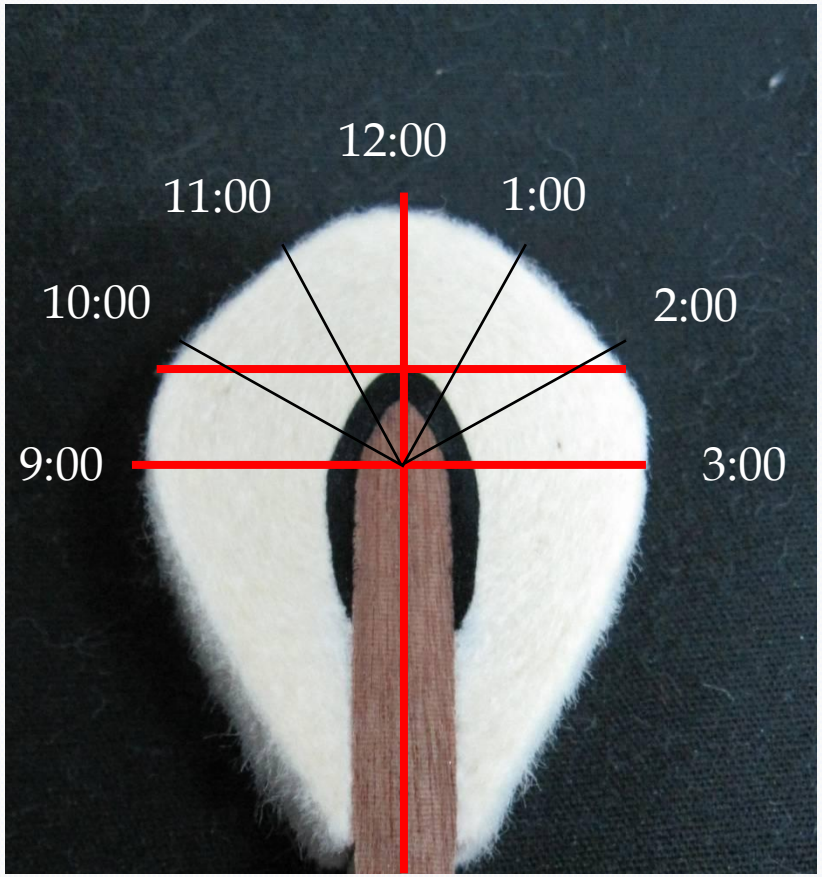


## Working steps while voicing

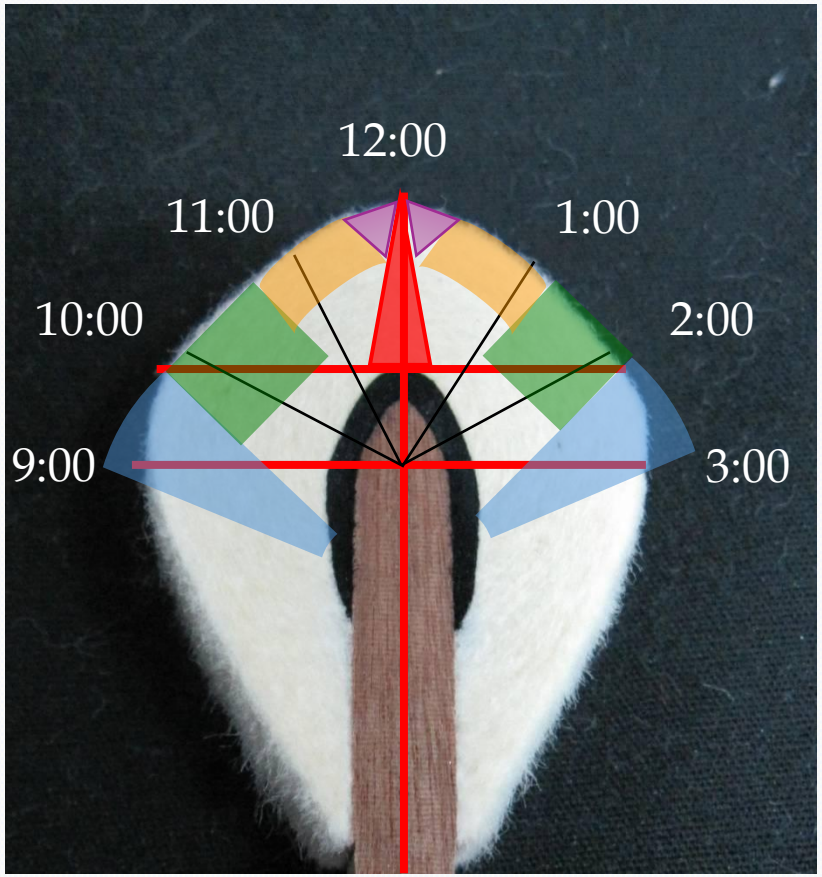
- Pre-voicing of the hammerheads
- Sanding | forming | ironing of the hammerheads
- Aligning | setting | leveling the strings
- Final tuning and regulation works
- Travelling hammers and aligning to strings
- File and straightening the hammers to strings
- Final section voicing
- Final balance voicing
- Balance shift pedal (left pedal) position

## Working goals trough voicing

- Increase and open dynamic range
- Tone length (sustain)
- Culture of touch | pianistic play style
- Balance in line
- Balance in tone
- Balance and characteristics of shift pedal

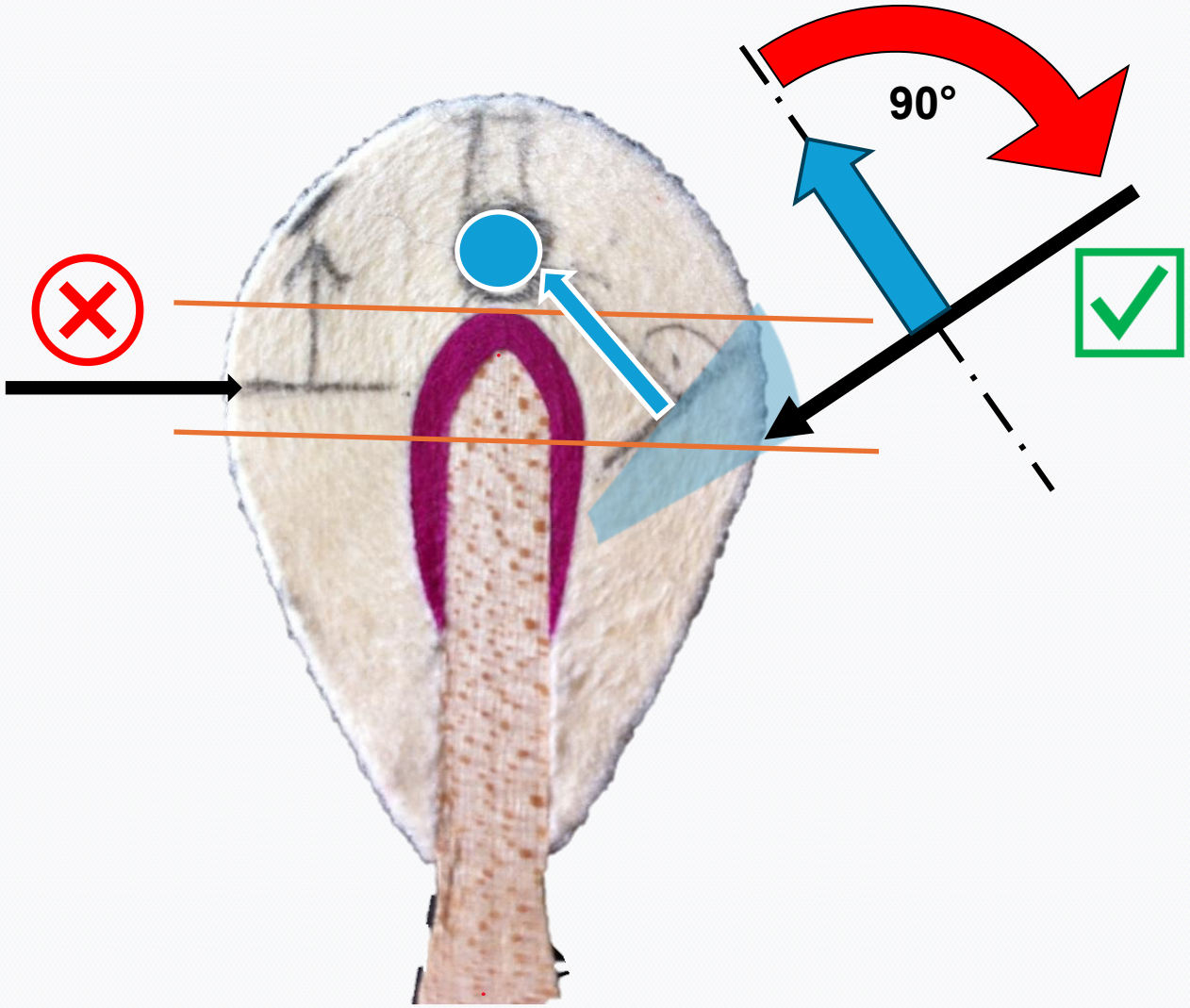



- Central energy line
- At a right angle highest cheek area
- At a right angle thereby top edge of underlay felt
- Remaining distribution like hours on a clock




- Purple = Pianissimo wedge
- Orange = Piano wedge
- Green = Forte wedge
- Blue = Fortissimo wedge
- Red = do not touch!!!

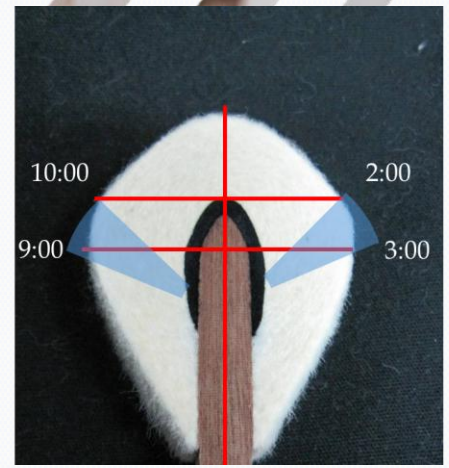



90°



**What to do?**

- Pre- needleing in the fortissimo wedge
- Deep stitches to just before the underfelt
- As a result the hammerhead looses any tension, the tension is transferred to the forte wedge
- Therefore, this first step of pre-voicing is advisable in almost every case




**How to do?**

- Clambing hammerheads (if should be done before gluing the hammerheads)
- Needle holder (voicing tool) with 2 or 3 needles, needle sizes 0,6- 0,8mm approx 9mm long
- Always work from the bottom up
- Do not stitch into underfelt or wooden core
- Your working zone is the fortissimo wedge, the areas between 9:00 and 10:00 and 2:00 and 3:00
- Bass 7-10 stiches horizontal
- Middle 5-10 stiches horizontal
- Treble 0-7 stiches vertical
- Pressing the needles is more effective and durable than hacking

**Tools**

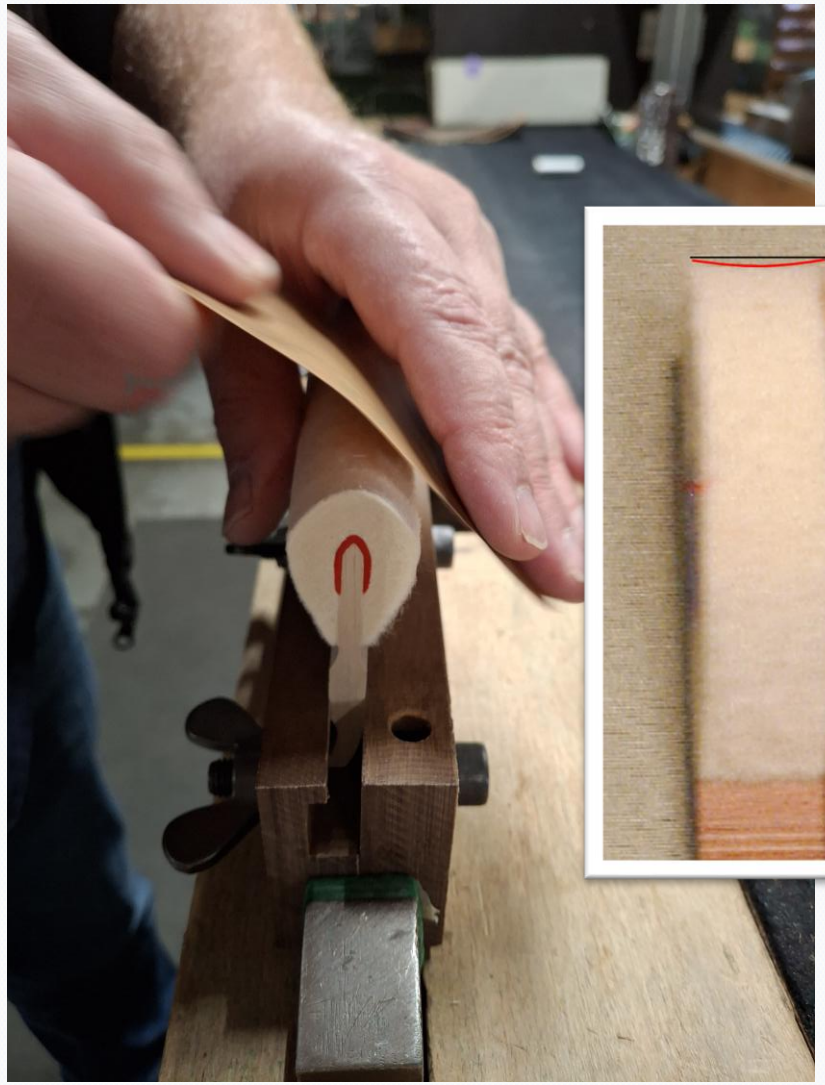
- Toning / voicing adjustable device for grand piano and upright hammerheads, Item number: 43-0005642
- Toning/voicing needle holder, professional type, with three needles, Item number: 43-0003270
- Voicing needle no.7 nickel plated Ø 0,60 mm bag with 25 pieces, Item number: 43-0003426

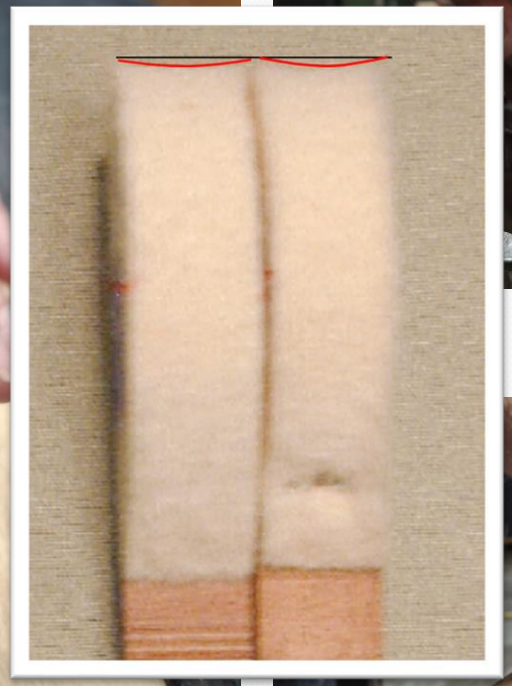


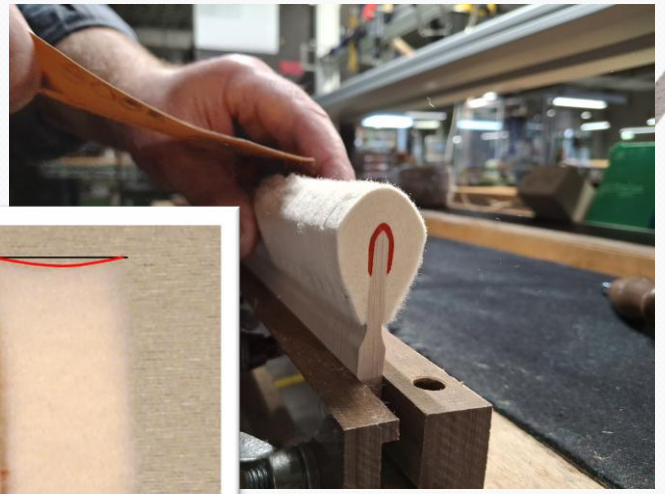


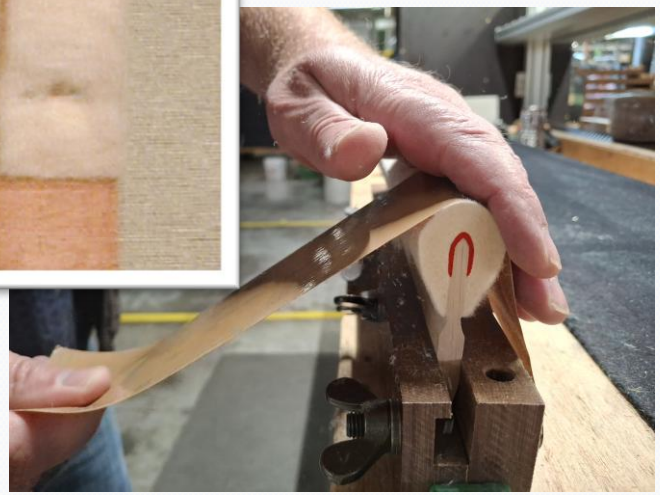


- This works can be done before gluing of the hammers









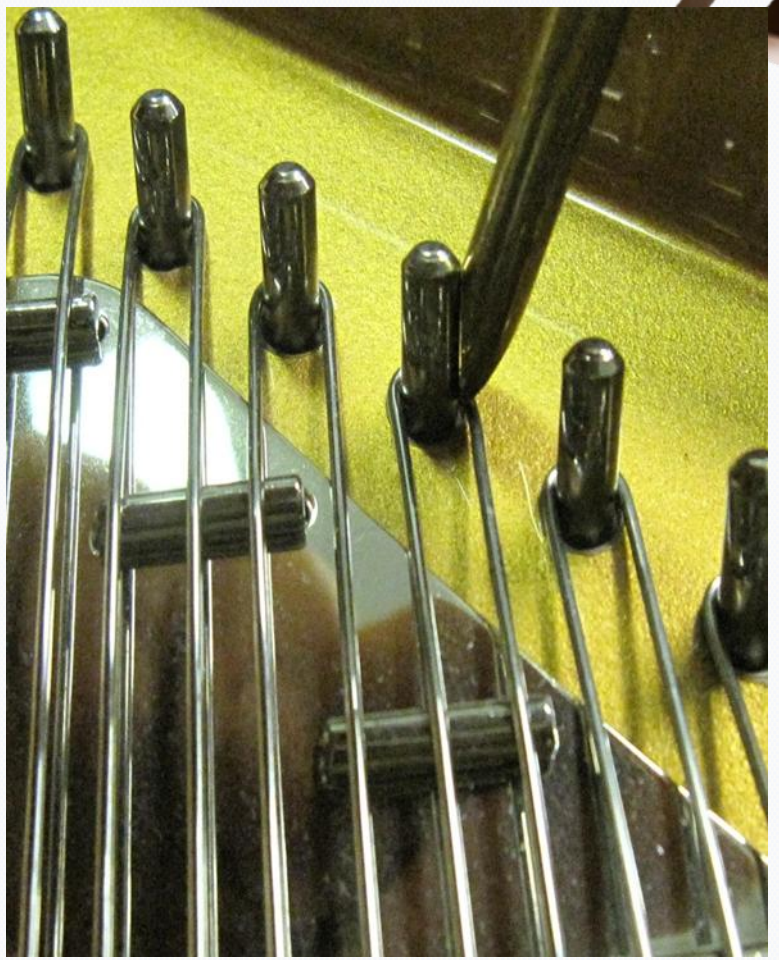


- This works can be done before gluing of the hammers

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# Sanding | forming | ironing surface of hammerheads



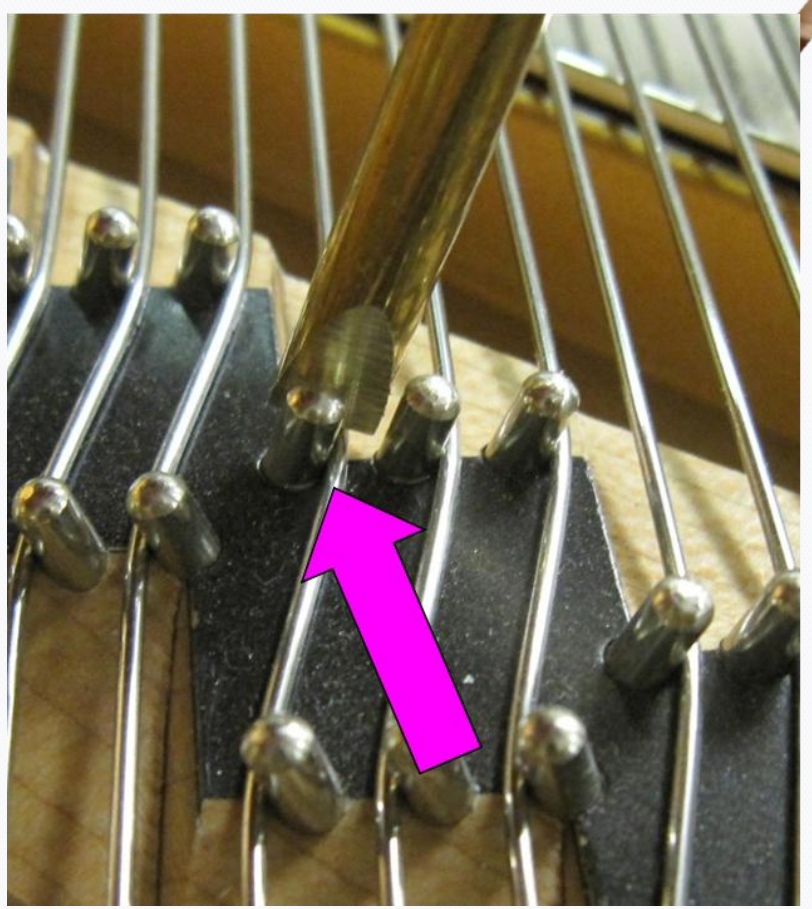


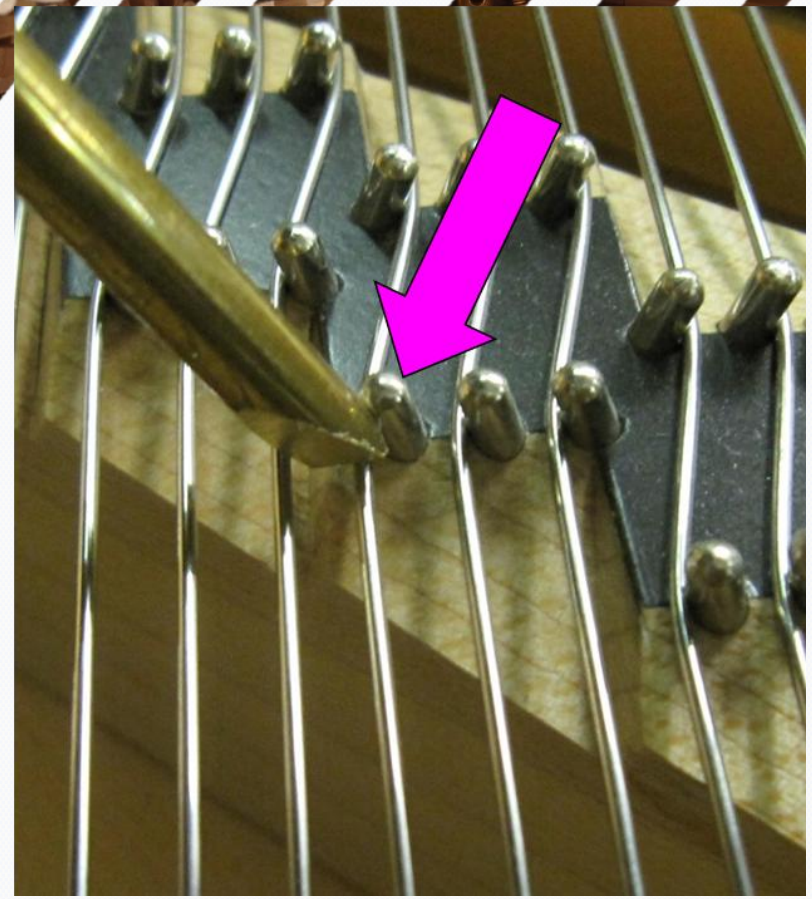
- Tuning is on 440 Hz or more
- Work from back to front,
  frame pin > bridge pins > agrafe
- Carefully tap the string down onto the pin
- Work with a brass chisel and light hammer as not to damage the string

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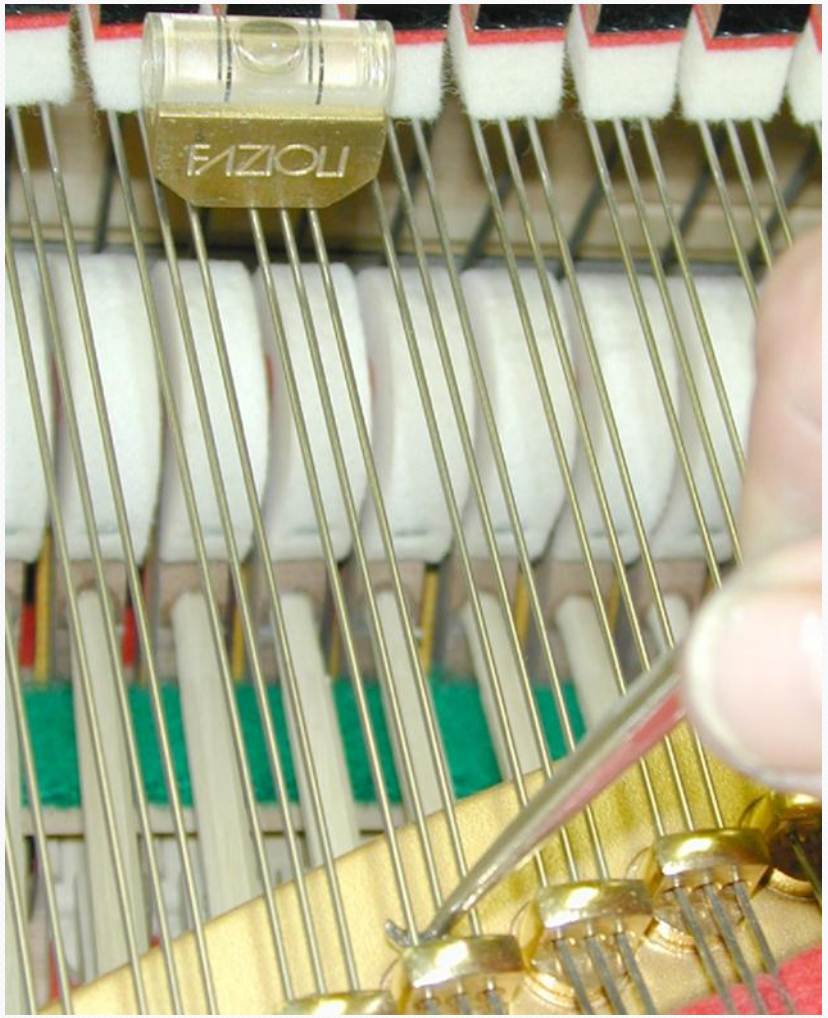
# Setting strings on frame pin

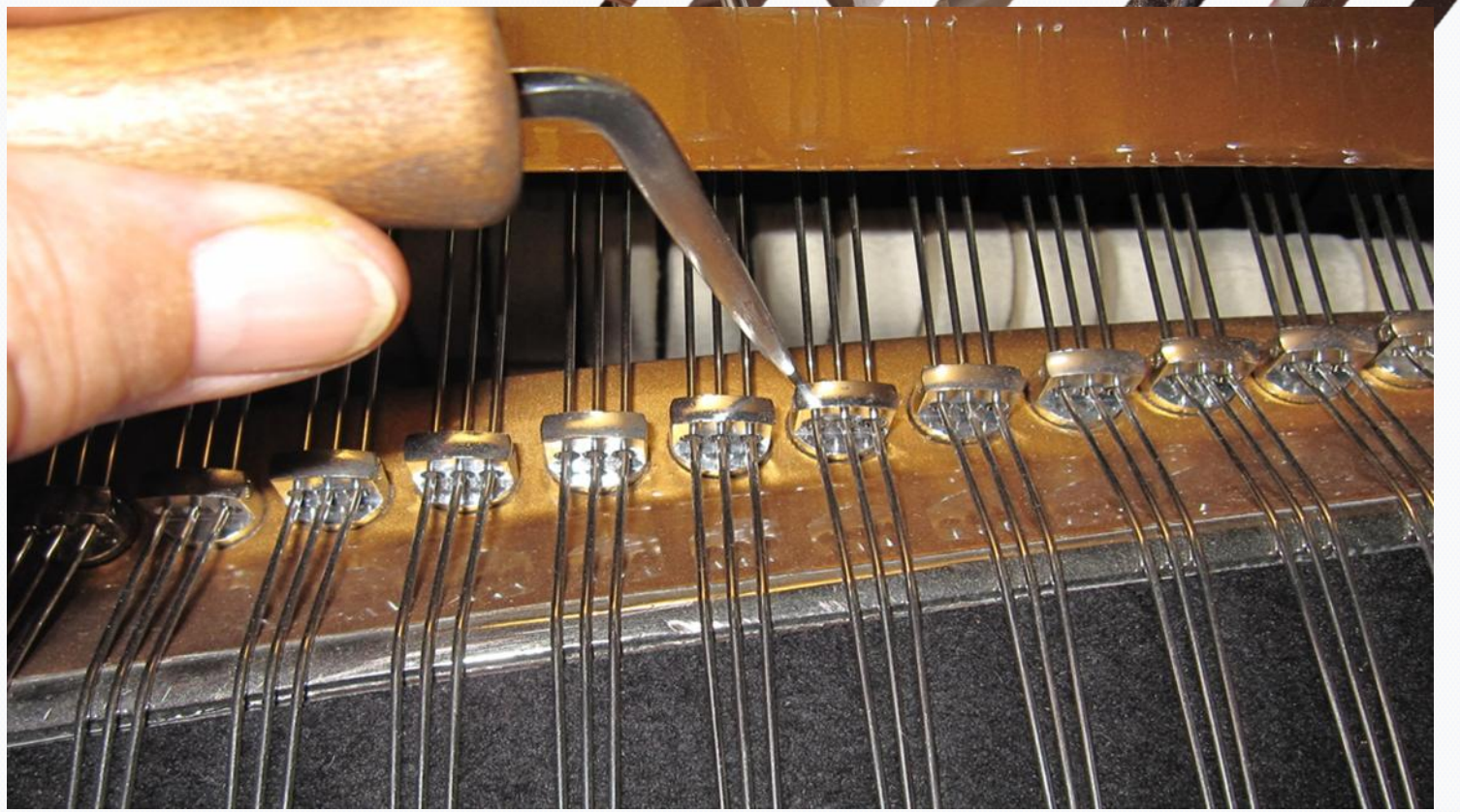
- Carefully tap the string down onto the bridge pin
- First the rear, then the front pin
- Same tools as before
- Work carefully so that the string does not press into the wood of the bridge (this could lead to noises)
- If unsure, knock on the inside
- Important: clearly defined string supports
- The fuller the strings rest on their limit point, the more powerfull is their sound





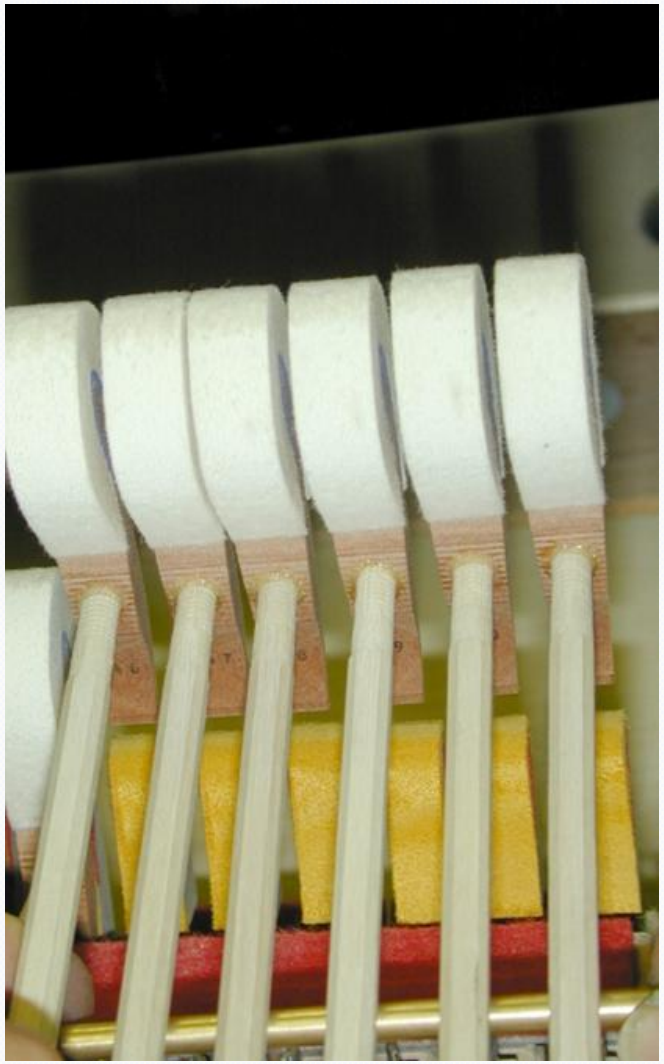
➢ **Notice:** the best hammerhead can only perform the function that was previously programmed into the instrument

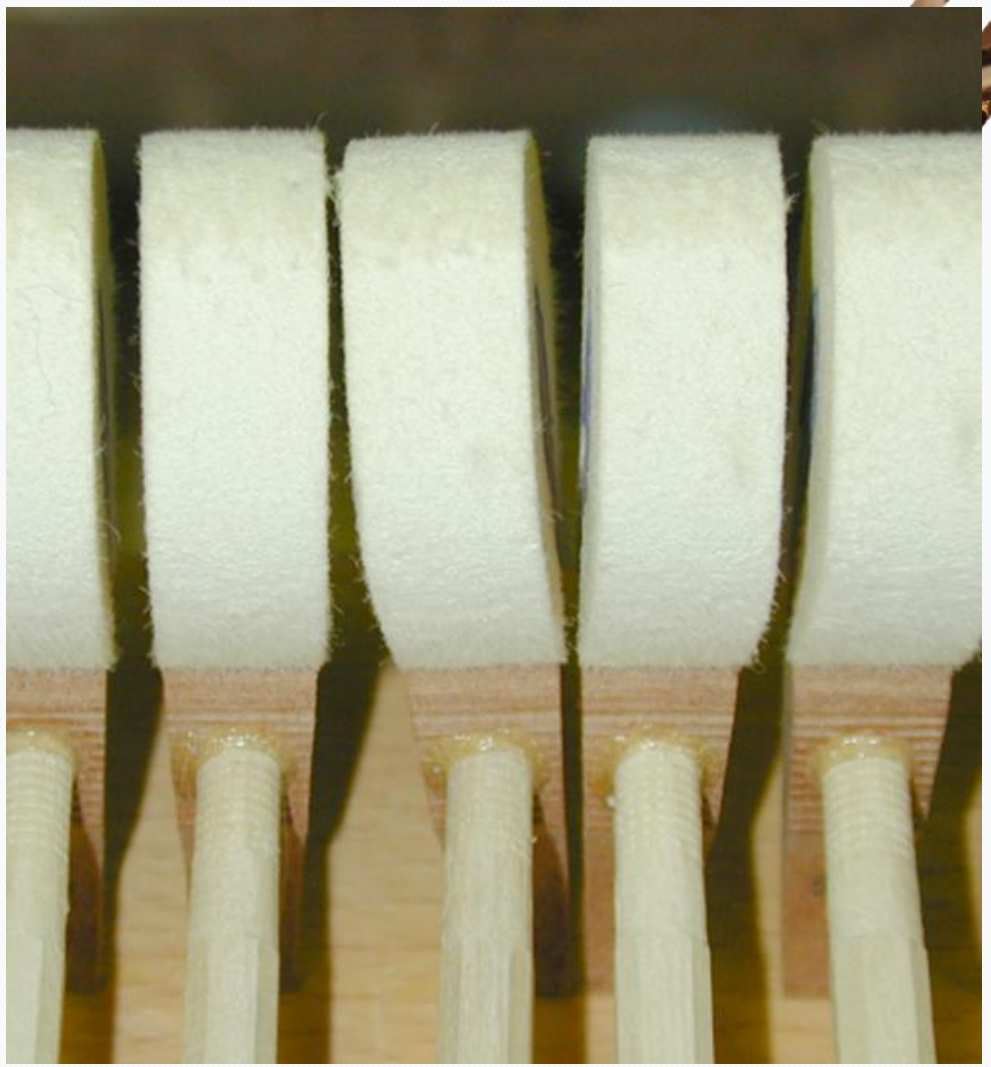


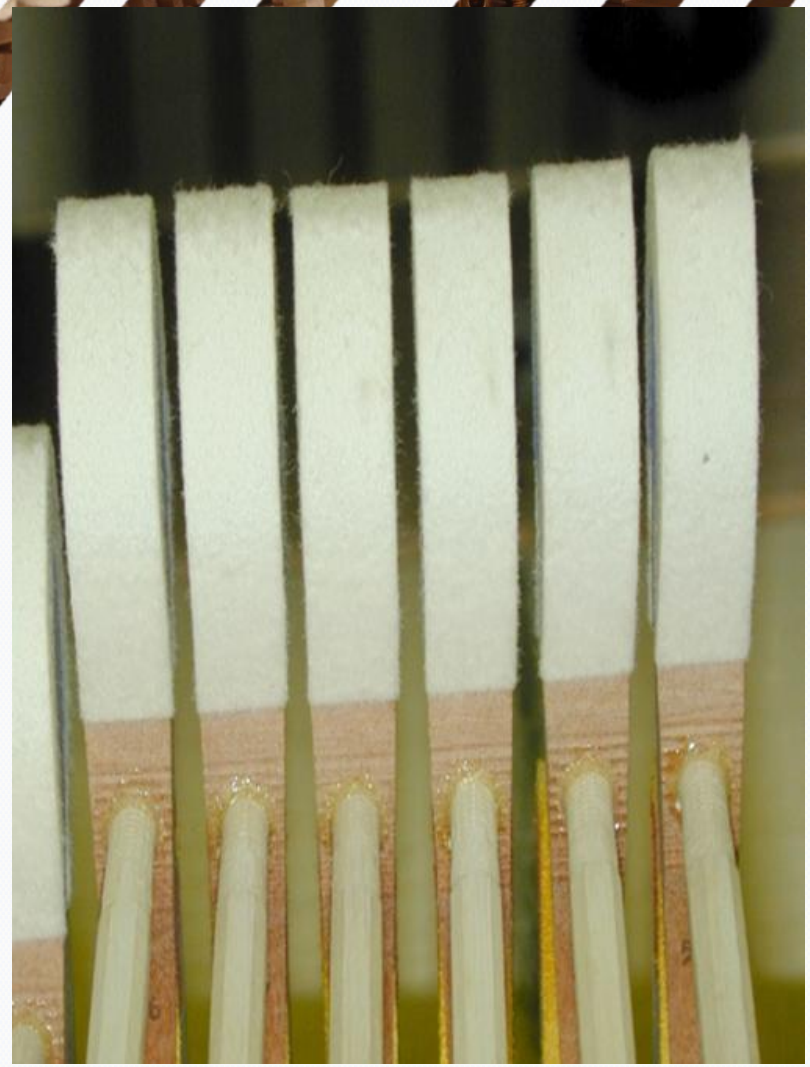



- Raising all strings and adjustment to a uniform level in agrafe (tis is not the same as spreading the string after stringing)
- Finally align the string choirs in treble
- Be sure to perform when the instrument is on final tuning level

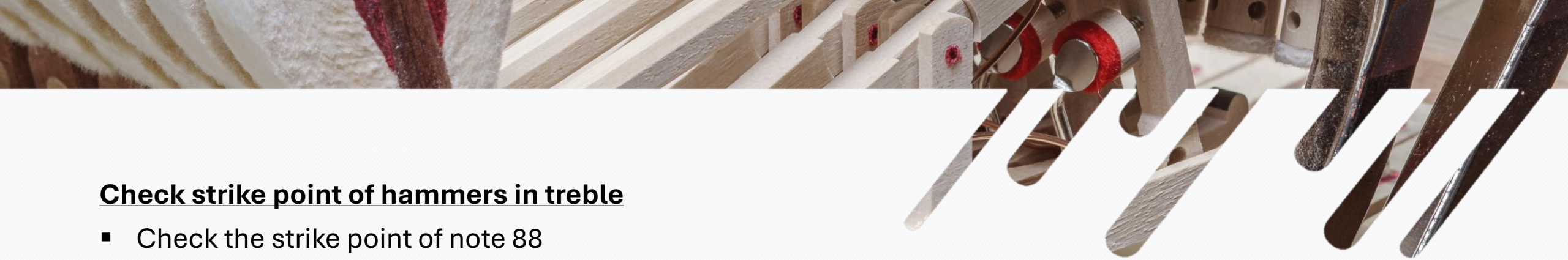








- Now all hammers must be aligned on the strings again

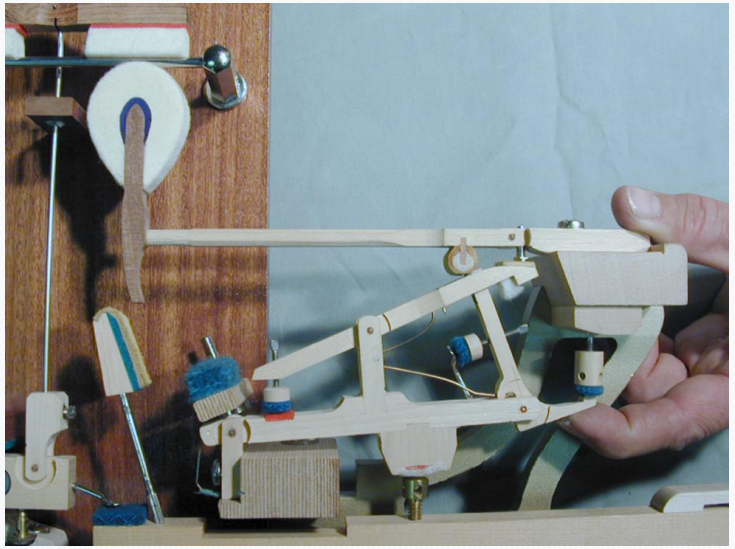


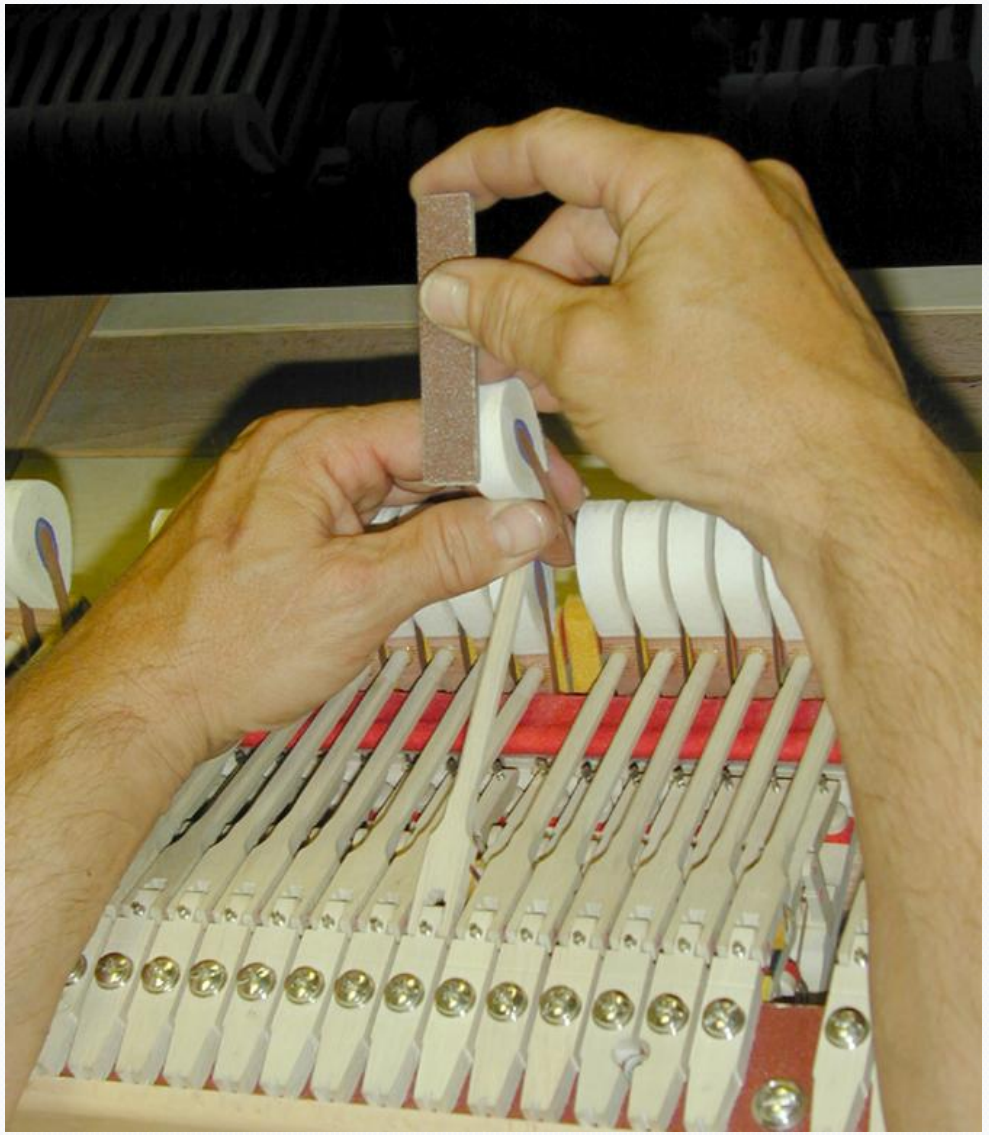
**Check strike point of hammers in treble**

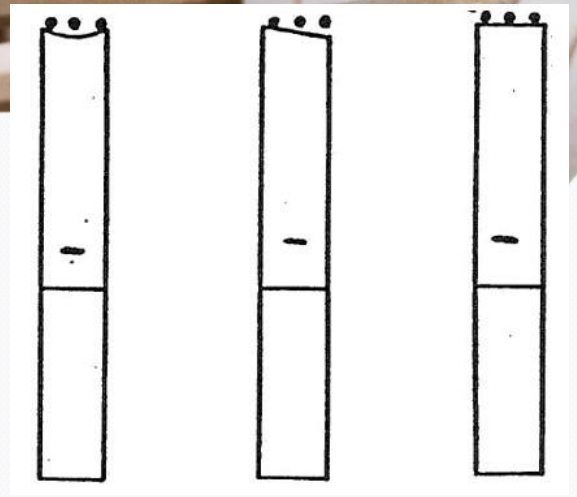
- Check the strike point of note 88
- Slide keyframe in and out to find best spot
- The treble section of many pianos can be improved by re-setting the strike point from notes 69 through 88
- Correct if necessary

**Check straightness of all hammerheads**

- Can be done by pulling up with a hook and plucking
- Can be done by lifting the jack with finger and plucking
- Can be done by feeling and hearing with a light tapping
- To work effectively, mark with chalk on the key what needs correction







Crown hollow | Crown crooked | Crown okay

**File hammers and straighten to strings**

- Due to the unevenness of the string chorus and the impacts of travelling the hammers, the surface of the hammerhead must be adjusted
- This is done by fine filing with a hammer file
- It‘s up to you to decide about the tools you prefer
- We have prepaired some for you, that you can take with you

**Mid-voicing**

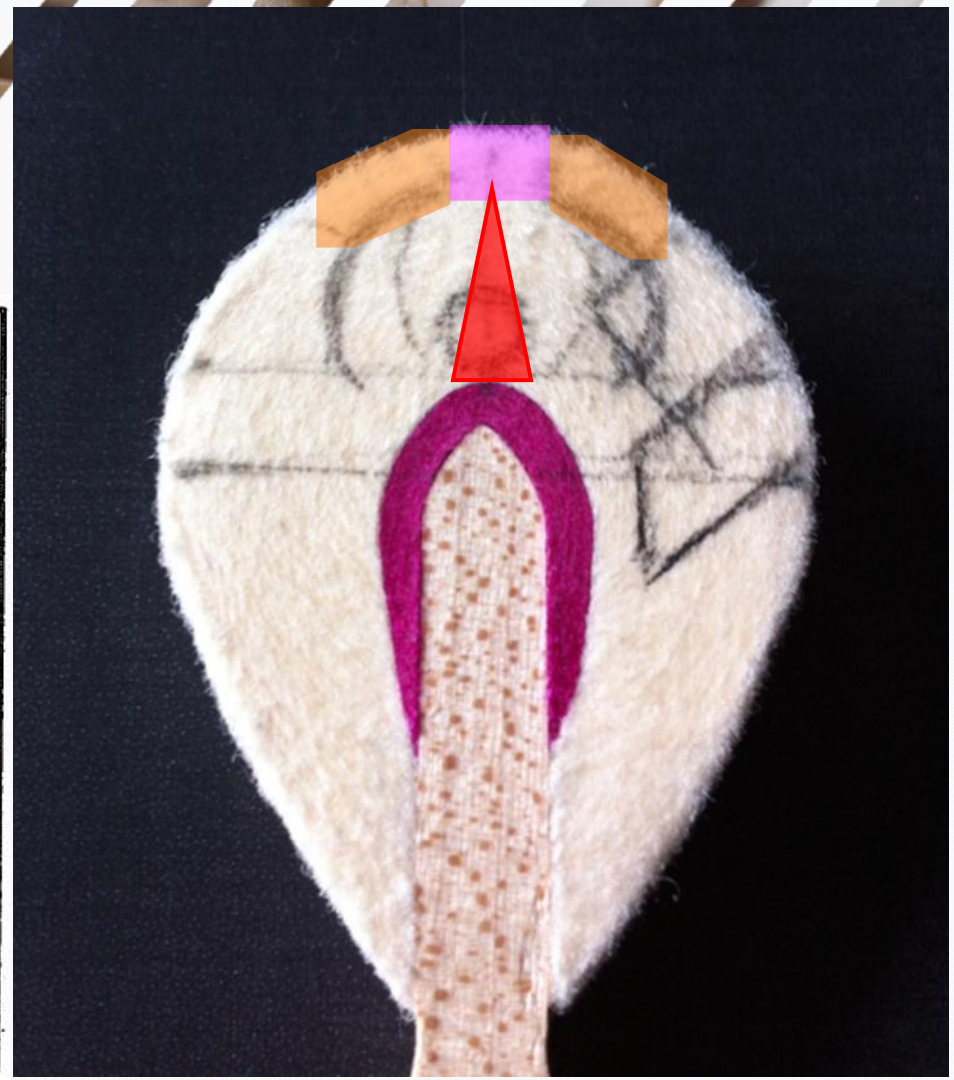
- Play up and down the keyboard chromatically
- Hear for sections, that have different characteristics, loudness or feeling
- Work out in green (forte) and orange (piano) wedge area if to hard
- This should help to create a balanced sound dynamic
- If you feel unsure how deep you can work prepare different tools with different needle length
- Be cautious but not fearful when stiching
- If a note or passage has become too soft, it can be corrected by lightly sanding
- Work out a balanced sound spectrum and volume level across the whole keyboard
- Accept the characteristics of the instrument



## Compensate final balance

- Find single notes with an other character or loudness
- Work with a short single needle to compensate
- Work in piano (orange) and pianissimo (purple) wedge
- Remember the **„red zone“** and do not work to deep at the crown!
- Play up and down the keyboard chromatically and listen for every uneven note
- Light shallow surface needling at or just under the crown will even out the offenders
- This is easily overdone, so proceed slowly
- Work between the string strokes, not directly inside
- Stich depth at the crown max. 2 mm





**Tool tip:**

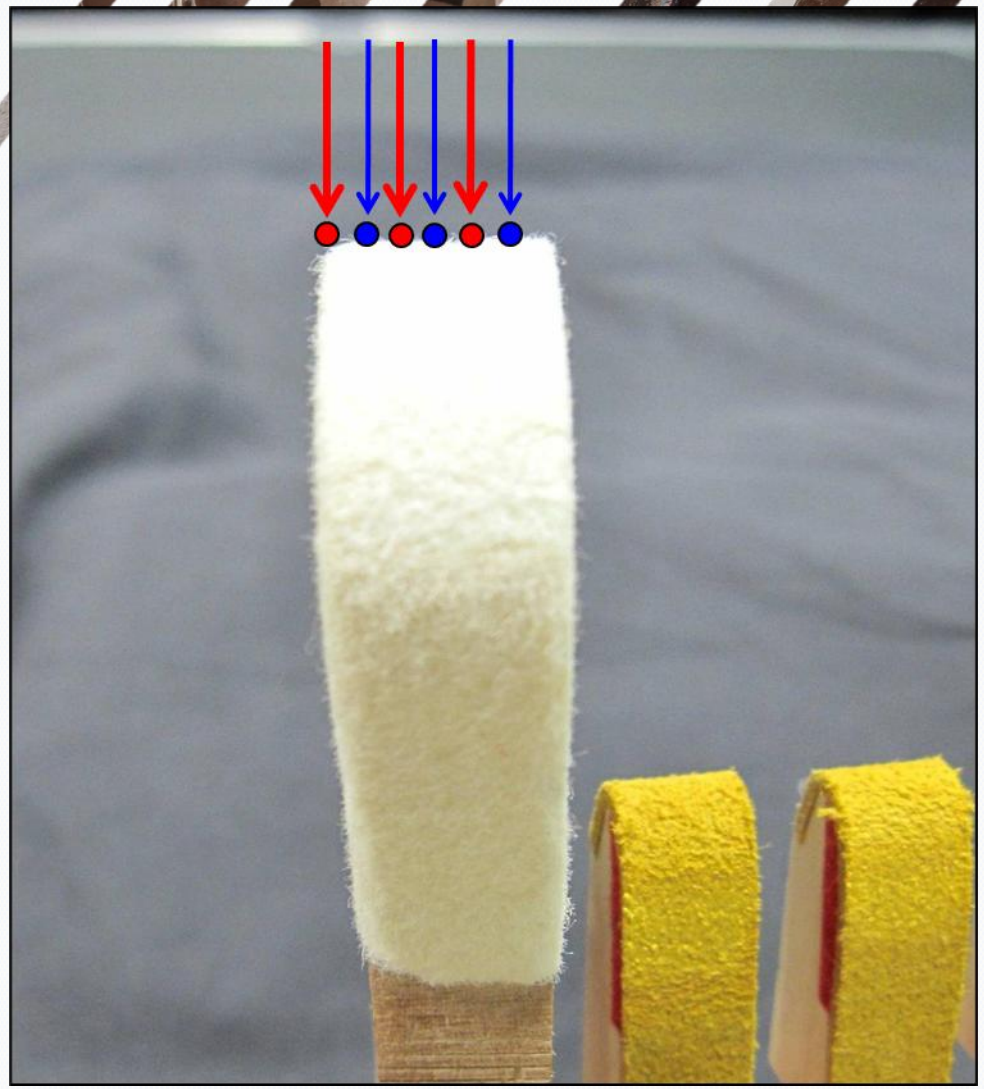
- Toning/voicing needle holder BDK, solid brass, for 4 needles
  Item number: 43-0003269
- Toning/voicing needle holder, Chop Stick Item number: 43-0003271

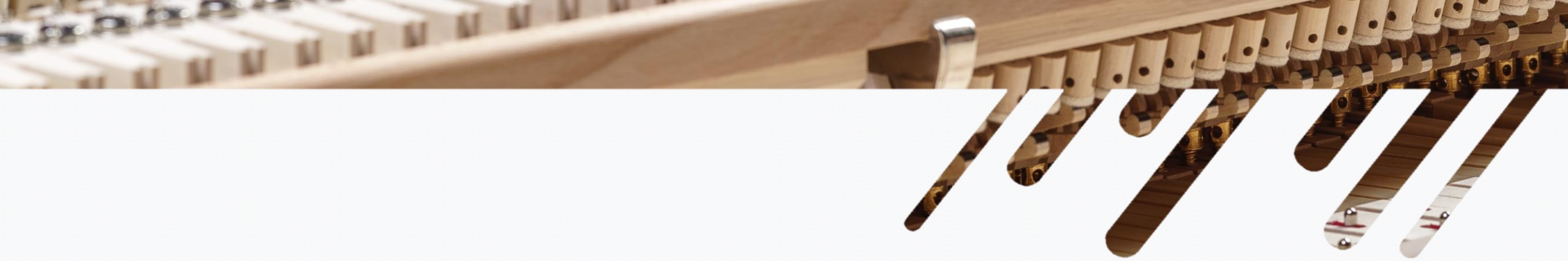
**Shift pedal voicing**

- Finally, adjust the shift pedal correctly
- Play up and down the keyboard chromatically with padel pressed and listen on uniformity
- Work with particular care, as the main voicing may be affected
- Always work only on the strings directly affected
- Then connect with the main voicing by checking balance without the pedal

**Expert tip:**

➢ Slightly breaking the left edge of the hammer crown with a fine sandpaper file can make this work much easier and can prevent noises





# Thank you for your interest

# Let the Perfect Hammer Sound!

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